TASK 2: CHATBOT FOR FAQs

A Python NLP Project Using TF-IDF and Cosine Similarity

Project Overview

This project implements a simple FAQ chatbot for a college help desk. It stores frequently asked questions and answers, cleans and preprocesses text, converts questions into TF-IDF vectors, and finds the closest FAQ using cosine similarity. The chatbot returns the best matching answer or asks the user to rephrase when confidence is low.

Technology Used

Python 3 • scikit-learn • Regular Expressions • TF-IDF • Cosine Similarity • Command-line Chat UI

Objective

To build a beginner-friendly chatbot that answers common college-related questions by matching a user's question with a stored FAQ.

1. Problem Statement

Students often ask repeated questions about admissions, fees, class timings, attendance, examinations, and library services. A FAQ chatbot can provide quick answers to these common questions without requiring a person to respond each time.

2. System Workflow

Step	Description
1. Collect FAQs	Prepare a list of questions and answers for the college help desk.
2. Preprocess	Convert text to lowercase, remove punctuation, and normalize extra spaces.
3. Vectorize	Use TF-IDF to convert FAQ questions into numerical vectors.
4. Match	Use cosine similarity to compare the user's question with stored questions.
5. Respond	Return the answer for the most similar FAQ if the score passes a threshold.

3. Sample FAQs

Q: What are the college working hours?
A: The college working hours are generally 9:00 AM to 4:00 PM. Please confirm with the college office.

Q: How can I pay my fees?
A: You can pay fees through the college's official payment portal or by contacting the accounts office.

Q: Where can I get my timetable?
A: The timetable is usually shared by your department or published on the student portal.

Q: What is the attendance requirement?
A: Attendance requirements depend on your college rules. Check the student handbook or contact your department.

Q: When are examinations conducted?
A: Examination dates are announced by the examination branch through official notices.

Q: How can I get a library card?
A: Contact the library desk with your student ID to learn the registration process.

Q: How do I contact the college office?
A: Use the phone number or email listed on the college's official website.

Q: Where can I find course materials?
A: Check your learning management system, department announcements, or ask your faculty.

Q: How can I reset my student portal password?
A: Use the portal's password-reset option or contact the IT support desk.

Q: Who should I contact for a technical issue?
A: Contact the college IT help desk or your department coordinator.

4. Complete Python Source Code

```
# Task 2: FAQ Chatbot
# Install dependency: pip install scikit-learn

import re
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

faqs = [
    {
        "question": "What are the college working hours?",
        "answer": "The college working hours are generally 9:00 AM to 4:00 PM. Please confirm with the college office."
    },
    {
        "question": "How can I pay my fees?",
        "answer": "You can pay fees through the college's official payment portal or by contacting the accounts office."
    },
    {
        "question": "Where can I get my timetable?",
        "answer": "The timetable is usually shared by your department or published on the student portal."
    },
    {
        "question": "What is the attendance requirement?",
        "answer": "Attendance requirements depend on your college rules. Check the student handbook or contact your depart
    },
    {
        "question": "When are examinations conducted?",
        "answer": "Examination dates are announced by the examination branch through official notices."
    },
    {
        "question": "How can I get a library card?",
        "answer": "Contact the library desk with your student ID to learn the registration process."
    },
    {
        "question": "How do I contact the college office?",
        "answer": "Use the phone number or email listed on the college's official website."
    },
    {
        "question": "Where can I find course materials?",
        "answer": "Check your learning management system, department announcements, or ask your faculty."
    },
    {
        "question": "How can I reset my student portal password?",
        "answer": "Use the portal's password-reset option or contact the IT support desk."
    },
    {
        "question": "Who should I contact for a technical issue?",
        "answer": "Contact the college IT help desk or your department coordinator."
    }
]

def preprocess(text):
    """Lowercase text, remove punctuation, and normalize spaces."""
    text = text.lower()
    text = re.sub(r"[^a-z0-9\s]", " ", text)
    return re.sub(r"\s+", " ", text).strip()

questions = [preprocess(item["question"]) for item in faqs]

# TF-IDF turns text into numerical vectors.
vectorizer = TfidfVectorizer(ngram_range=(1, 2))
faq_vectors = vectorizer.fit_transform(questions)

def get_answer(user_question, threshold=0.18):
    cleaned = preprocess(user_question)
    if not cleaned:
        return "Please enter a question."

    user_vector = vectorizer.transform([cleaned])
    scores = cosine_similarity(user_vector, faq_vectors)[0]
    best_index = scores.argmax()
    best_score = scores[best_index]

    if best_score < threshold:
        return ("Sorry, I couldn't find a close FAQ match. "
                "Please rephrase your question or contact the college office.")

    return faqs[best_index]["answer"]
```

```
def main():
    print("College FAQ Chatbot")
    print("Ask a question. Type 'exit' to stop.")
    while True:
        user_question = input("You: ").strip()
        if user_question.lower() in {"exit", "quit", "bye"}:
            print("Bot: Goodbye! Have a nice day.")
            break
        print("Bot:", get_answer(user_question))

if __name__ == "__main__":
    main()
```

5. How to Run the Project

1. Install Python 3 on your computer.

2. Open Command Prompt or Terminal.

3. Install the required library by running: pip install scikit-learn

4. Save the source code as faq_chatbot.py.

5. Run the program using: python faq_chatbot.py

6. Type a question such as “How can I pay my fees?” and read the chatbot response.

7. Type exit to close the chatbot.

6. Sample Output

```
College FAQ Chatbot
Ask a question. Type 'exit' to stop.
You: How do I pay my college fees?
Bot: You can pay fees through the college's official payment portal or by contacting the accounts office.
You: Where is my timetable?
Bot: The timetable is usually shared by your department or published on the student portal.
You: Tell me a joke
Bot: Sorry, I couldn't find a close FAQ match. Please rephrase your question or contact the college office.
You: exit
Bot: Goodbye! Have a nice day.
```

7. Result

The FAQ chatbot was designed using Python, text preprocessing, TF-IDF vectorization, and cosine similarity. It compares a user's question with stored FAQ questions and returns the most relevant answer when the similarity score is sufficient. The FAQ list and similarity threshold can be adjusted for a specific product, service, or institution.

8. Limitations and Future Enhancements

This is a small, offline prototype and only answers questions represented in its FAQ dataset. Future improvements could include a web-based chat interface, a larger FAQ database, stemming or lemmatization, intent classification, and connection to an approved college knowledge base. The sample college timings and policies are illustrative; replace them with verified information before real use.